GG-M-408b
AUGUST 6, 1958

SUPERSEDING
Int. Fed. Spec. GG–M–00408a (GSA–FSS)
June 18, 1956 and
Fed. Spec. GG–M–408
August 5, 1946
(See section 6.)

FEDERAL SPECIFICATION

MIRROR, HEADBAND; AND PROTECTOR, HEAD MIRROR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers a mirror and mirror protector for use with the mirror to be attached to headbands as used in the medical and dental profession.

1.2 Classification.

1.2.1 *Classes.*—The mirror and the mirror protector shall be of the following classes, as specified (see 6.1).

Class 1.—Mirror.

Class 2.—Mirror protector.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG–I–526—Instrument, Dental and Surgical; General Specifications.

PPP–B–566 — Boxes, Folding, Paperboard.

PPP–B–585—Boxes, Wood, Wirebound.

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–117—Bags, Interior Packaging.

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MIL–B–10377—Box, Wood, Cleated, Veneer, Paper, Overlaid.

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.*—The glass shall be of the best quality, polished, free from waviness and other imperfections, and of such composition as to withstand prolonged usage without its surface becoming dulled.

3.1.2 *Brass.*—The brass shall be in accordance with Federal Specification GG–I–526, class 4, composition N.

3.1.3 *Other materials.*—Shall be as hereinafter specified.

3.2 Design and construction.

3.2.1 *Class 1, mirror.*—Head mirrors shall consist of a mirror, frame, and back, as shown in figure 1. The back shall have a ball post for attaching mirror to headband.

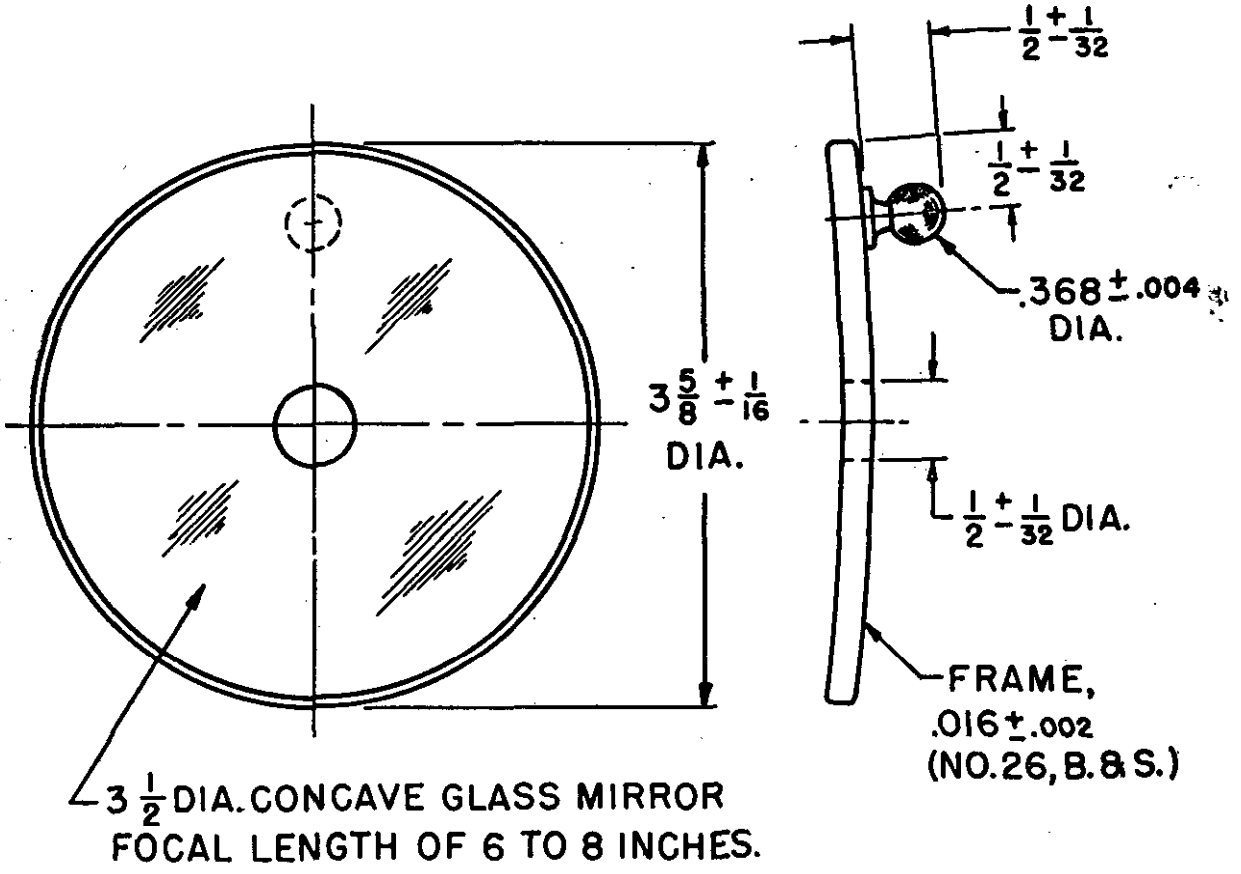


FIGURE 1.

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3.2.1.1 *Mirror proper.*—The mirror shall be made of thin glass, circular in form, 3.5 inches in diameter, and sufficiently concave to give a focal length which shall lie within a minimum of 6 inches and a maximum of 8 inches. The back side of the mirror shall be properly silvered, then heavily electroplated with copper, and finished with a water-resistant baked-on coating of black enamel which shall be carried over onto the side edge. The silvering shall be firmly adherent and shall be capable of passing the boiling and heating test specified in 4.3.1. The mirror shall reflect a clear undistorted image. There shall be a hole 0.5 inch in diameter extending through the exact center of the mirror.

3.2.1.2 *Frame and back.* — The mirror frame shall be fabricated of a thin sheet of brass, or other suitable metal, which shall be attached by spinning the metal well up over the edge of the glass. Water-resistant cardboard, plain cork, rubber, or other suitable water-resistant cushioning material, adequate to protect the back of the mirror from injury, shall be placed between the frame and the mirror. In the exact center there shall be a circular opening 0.5 inch in diameter extending through the metal frame and backing exactly over the hole in the mirror.

3.2.1.3 *Ball post.*—The ball post shall be of brass or other suitable alloy or metal, chromium over nickel plated. The ball post shall be firmly attached to the outside surface of the metal frame and near its periphery, by riveting over a brass washer. The washer shall be located between the peaned portion of the knob and the inside surface of the metal frame. The ball post shall be constructed as shown in figure 1.

3.2.1.4 *Weight.*—Both the metal back and glass shall be as light as is consistent with strength and durability.

3.2.2 *Class 2, mirror protector.*—The mirror protector shall be in accordance with figure 2. The protector shall be fabricated of brass or carbon steel with edges that shall be smoothly ground leaving no splinters, protruding points, or "wire edge". The screw and nut shall be fabricated of brass.

3.3 **Finish.**—The ball post shall be chromium over nickel plated. The frame and back shall be finished with a baked-on carbon black enamel. The protector shall have a chemical black finish.

3.4 **Heat resistance.**—The assembled headband mirror shall show no evidence of corrosion, scumming, or dulling of the surface, nor shall the mirror loosen in the frame, when tested as described in paragraph 4.3; nor shall there be visual evidence of flaking, blistering, peeling, or other injury to the silvered surface of the mirror, as a result of the test.

3.5 **Marking.**—Each mirror shall be legibly and permanently marked with the name or registered trade-mark of the manufacturer.

3.6 **Workmanship.** — The workmanship shall be first class throughout. Head mirrors and protectors shall be free from any avoidable defects which will detract from its appearance or may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**

4.1.1 *For visual examination.*—Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.1.2 *For tests.*—Sample size for each test specified shall be as indicated for inspection L–2 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero for all sample sizes, and a minimum of four samples shall be taken regardless of lot size. The unit of product for inspection purposes shall be one mirror.

4.2 **Inspection.**—The inspector shall ascertain that the mirrors have been inspected and accepted in accordance with specification requirements.

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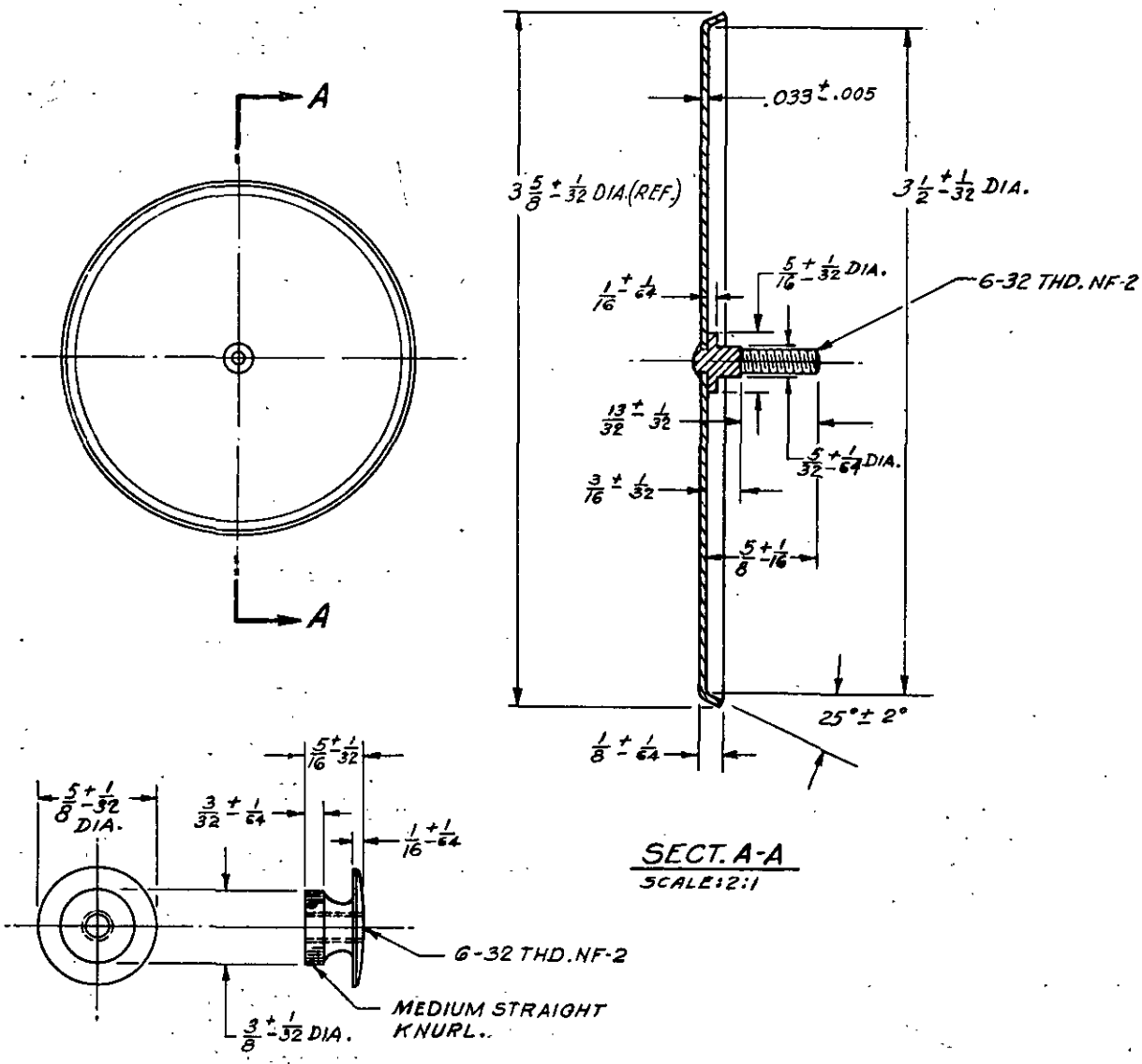


FIGURE 2.

4.3 Tests.

4.3.1 *Boiling and heating test.*—The head mirrors selected for test shall be thoroughly cleansed with soap and water and then rinsed thoroughly with distilled water. After cleaning, the head mirrors shall be totally submerged in distilled water, in glass apparatus, and boiled continuously for 24 hours. Boiling distilled water shall be added as necessary to replace water lost during the test. Following completion of boil test, the mirrors shall be dried for 1 hour at 100° C.

4.3.2 *Focal length determination.* — The focal length of the mirror shall be determined by using a collimated light source, or a light source at infinite distance (minimum of 50 feet) from the mirror to be tested. The mirror shall be positioned so that the reflecting face is nearly perpendicular to the light source. A white card (or sheet of paper) shall be placed in front of the mirror in such a position so as not to block the light source from the mirror face, but to be in the focal plane of the mirror. The card shall be moved away from, or towards, the mirror until the

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sharpest image (or brightest spot) is found. The distance from the card at this point to the mirror face shall be considered as the focal length of the mirror.

4.4 Supplier's responsibilities for inspection.—The supplier shall perform examinations and tests set forth in this specification and substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities, or shall arrange for test facilities acceptable to the Government to be used for executing specified prescribed tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government upon the Government's request, at any time, or from time to time, for a period of two years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

The levels of preservation, packaging and packing protection used are defined in Federal Standard No. 102 (for civil agencies only).

5.1 Packaging.

5.1.1 *Unit of issue.*

5.1.1.1 *Stock No. 6515–347–5200.* — One mirror, as specified, constitutes one unit.

5.1.1.2 *Stock No. 6515–347–5320.* — One protector, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit, intermediate and exterior container as applicable for the required level of protection specified in the procurement document.

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*

5.1.3.1.1 *Stock No. 6515–347–5200.*—Each unit shall be wrapped in tissue and packaged in a carton conforming to Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling. Carton shall be further packaged in a bag conforming to Military Specification MIL–B–117, type II, class d.

5.1.3.1.2 *Stock No. 6515–347–5320.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

TABLE I.—*Packaging quantities*

Federal stock number	Unit package	Intermediate package	Shipping container
6515–347–5200	1 unit	12 units	360 units
6515–347–5320	1 unit	6 units	288 units

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5.1.4 *Level B.*

5.1.4.1 *Unit package.*

5.1.4.1.1 *Stock No. 6515–347–5200.*—Each unit shall be wrapped in tissue and packaged in a carton conforming to Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4.1.2 *Stock No. 6515–347–5320.*—Package shall be as specified in 5.1.3.1.2.

5.1.4.2 *Intermediate package.*—Intermediate package shall be either a carton of suitable size and design, or a complete wrapping with kraft paper of suitable weight. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.5 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III, or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I, or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Unit package.*—Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.3.2 *Intermediate package.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.3 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Ordering data.** — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(*a*) Title, number and date of this specification.

(*b*) Classes required (see 1.2.1).

(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(1) Level B packaging is intended to provide economical but limited protection, and should be specified only when it is determined the mirrors will be held in covered storage approximately twelve months from date of initial packaging.

6.2 Storage surveillance.—Items packaged in accordance with level B requirements must be inspected to determine conditions when not used within the time period indicated in 6.1 (*c*) (1). Item not used within the time period specified must be repackaged in accordance with level B requirements in this specification or with level A requirements if storage beyond an additional year is anticipated.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid samples should be definitely stated, the specification to apply in all other respects.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

6.5 Transportation description.—Transportation description applicable to this item is:

Mirrors, not otherwise indexed by name.
Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Mirrors, not otherwise indexed, not bent, 120 united inches or under.
Motor volume minimum weight 12,000 pounds.

6.6 Intended use.—The mirror and mirror protector described in this specification shall be suitable for attachment to headbands, Federal Stock Numbers 6515–340–7700, Headbands, Mirror, Nylon and 6515–340–7900, Headbands, Mirror, Leather, Jefferson.

6.7 This specification covers the following items appearing in the Armed Services Medical Stock List:

6515–347–5200 — MIRROR, HEADBAND, 3½ Inch Diameter, ½ Inch Opening.

6515–347–5320—PROTECTOR, HEAD MIRROR, Metal.

6.8 Supersession data.—This specification supersedes Military Specification MIL–S–2267, dated 12 July 1950.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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